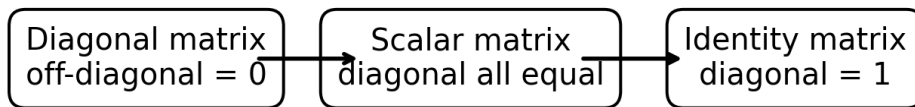


KUET MATHEMATICS - III

SECTION A CLASS NOTE

Math 2123 | Exam-oriented, source-grounded study edition



Memory ladder: Diagonal -> Scalar -> Identity

Core memory ladder used throughout Set A 1.

Matrix • Linear Systems • Eigen Analysis • Vector Kinematics • Work • Cayley-Hamilton

Credit: Made by 2223033

Prepared from the uploaded class notes, reference books, and the 2015-2025 Math 2123 question bank.

How to use this note

Exam strategy Section A has 4 sets in this analysis framework, but you answer 3. Sets A1-A3 are the safest core because they repeat most consistently. Set A4 is the flexible backup; however, its kinematics/work questions are often scoring when practiced.

Set	Main block	Best use
A1	Matrix fundamentals + short proofs	High-return theory + 10-15 mark matrix tasks
A2	Inverse + rank + consistency + dependence	Algorithmic, predictable, very exam-friendly
A3	Forms + eigenvalue/eigenvector + diagonalization	Long but structured; usually high marks
A4	Vector kinematics + work + Cayley-Hamilton	Backup set; formula-driven and scoring

বাংলা ব্যাখ্যার নিয়ম কঠিন ইংরেজি term-এর পাশে ছোট বাংলা অর্থ দেওয়া হয়েছে। কিন্তু formula, theorem name, matrix notation ইংরেজিতেই রাখা হয়েছে—যাতে exam script-এর language-এর সাথে মিল থাকে।

Source note Historical papers sometimes moved the same topic between Section A and Section B. This class note follows the A1-A4 topic map from the Question Bank Analysis, while the “Matching Question” tables tag the historical appearances regardless of old section placement.

Fast navigation

Set	What you must master	Memory hook
A1	Types, transpose, decomposition, orthogonal, idempotent/involutory/nilpotent	Mirror / sign / power
A2	ERO inverse, rank, consistency, homogeneous systems, dependence	D-R-I + R-R-n
A3	REF/RREF/normal, eigen, diagonalization	Pivot path + D-S-V
A4	v-a, TNB, curvature/torsion, work, CH	$v \rightarrow T \rightarrow N \rightarrow B + p(\lambda) \rightarrow p(A)$

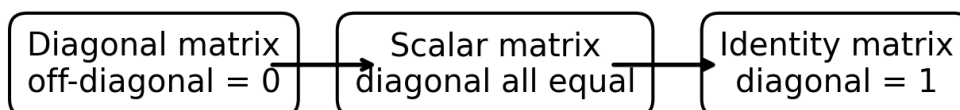
VERY HIGH PRIORITY

SET A1 — Matrix Fundamentals & Core Theorems

এই set-এ short definition, "define with example", ছোট proof, matrix classification এবং decomposition খুব বেশি আসে। লক্ষ্য: condition দেখে 3-5 সেকেন্ডে matrix type চিনতে পারা।

1. Matrix type master table

Type	Test / condition	বাংলা cue
Square	$m=n$	সারি = কলাম
Diagonal	$a_{ij}=0$ for $i \neq j$	মূল কর্ণ ছাড়া সব 0
Scalar	Diagonal + all diagonal entries equal	একই scalar বারবার
Identity	Scalar with diagonal 1	1 on diagonal
Upper triangular	$a_{ij}=0$ for $i > j$	কর্ণের নিচে 0
Lower triangular	$a_{ij}=0$ for $i < j$	কর্ণের উপরে 0
Symmetric	$A^T=A$	mirror করলে একই
Skew-symmetric	$A^T=-A$	mirror করলে sign বদলায়; diagonal 0
Orthogonal	$A^T A = A A^T = I$	transpose = inverse
Singular	$\det(A)=0$	inverse নেই
Non-singular	$\det(A) \neq 0$	inverse আছে
Idempotent	$A^2=A$	দুইবার করলেও একই
Involutory	$A^2=I$	নিজেই নিজের inverse
Nilpotent	$A^k=0$ for some $k \geq 1$	কিছু power-এ zero
Hermitian	$A^\dagger=A$	complex symmetric-এর analogue
Skew-Hermitian	$A^\dagger=-A$	complex skew analogue
Elementary	I-এর উপর one elementary operation	one-operation matrix
Equivalent	$B=PAQ$, P, Q non-singular	row+column ops-এ পাওয়া
Trace	$\text{tr}(A)=\sum a_{ii}$	diagonal sum
Principal submatrix	same indexed rows and columns retained	একই row/column index set



Memory ladder: Diagonal -> Scalar -> Identity

Shortcut — D-S-I Identity $\subset$ Scalar $\subset$ Diagonal. তাই “identity কি diagonal?” — হ্যাঁ। “diagonal কি scalar?” — সবসময় নয়।

2. Symmetric vs skew-symmetric — mirror rule

$$\text{Symmetric: } A^T = A \quad | \quad \text{Skew-symmetric: } A^T = -A$$

Skew-symmetric হলে $a_{ii} = -a_{ii} \Rightarrow a_{ii} = 0$

মনে রাখুন Main diagonal-কে আয়না ভাবুন। Symmetric-এ আয়নার দুই পাশে সংখ্যা একই; skew-symmetric-এ magnitude একই কিন্তু sign বিপরীত।

Theorem: every real square matrix splits into symmetric + skew-symmetric parts

$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T) = P + Q$$

$P^T = P$ and $Q^T = -Q$

Proof idea: transpose P এবং Q নিলেই যথাক্রমে P ও -Q পাওয়া যায়। Uniqueness-এর জন্য $A = P + Q$ এবং $A^T = P - Q$ যোগ/বিয়োগ করলে $P = \frac{1}{2}(A + A^T)$, $Q = \frac{1}{2}(A - A^T)$ বাধ্যতামূলক।

Worked example

Let $A = \begin{bmatrix} 1 & 3 & -1 \\ 0 & 2 & 4 \\ 2 & -1 & 5 \end{bmatrix}$. Then:

$$P = \frac{1}{2}(A + A^T) = \begin{bmatrix} 1 & 3/2 & 1/2 \\ 3/2 & 2 & 3/2 \\ 1/2 & 3/2 & 5 \end{bmatrix}$$

$$Q = \frac{1}{2}(A - A^T) = \begin{bmatrix} 0 & 3/2 & -3/2 \\ -3/2 & 0 & 5/2 \\ 3/2 & -5/2 & 0 \end{bmatrix}$$

Check $P + Q = A$, $P^T = P$, $Q^T = -Q$. Exam-এ তিনটি check লিখলে solution সম্পূর্ণ ও convincing হয়।

3. Orthogonal matrix

$$A \text{ orthogonal} \Leftrightarrow A^T A = I \Leftrightarrow A A^T = I \Leftrightarrow A^{-1} = A^T$$

Geometric meaning: orthogonal transformation length এবং angle preserve করে। তাই columns (এবং rows) unit এবং mutually orthogonal হয়।

Exam test example

$$B = (1/3) \begin{bmatrix} 2 & 2 & 1 \\ -2 & 1 & 2 \\ 1 & -2 & 2 \end{bmatrix}$$

Column dot-products: each column has squared length $(4+4+1)/9=1$; distinct columns have dot product 0. Hence $B^T B = I$, so B is orthogonal.

Shortcut 3×3 matrix-এ full multiplication না করেও columns-এর length 1 এবং pairwise dot product 0 দেখালে $B^T B = I$ প্রমাণ হয়ে যায়।

4. Idempotent, involutory, nilpotent

Type	Power test	Immediate consequence / clue
Idempotent	$A^2=A$	$A^n=A$ for $n \geq 1$; eigenvalues 0 or 1
Involutory	$A^2=I$	$A^{-1}=A$; eigenvalues ± 1
Nilpotent	$A^k=0$	all eigenvalues 0; smallest such k = index

Nilpotent example: $N = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $N^2=0$, so index = 2. Involutory example: $S = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $S^2=I$.

Proof pattern: if A is involutory, show $\frac{1}{2}(I \pm A)$ are idempotent

$$P = \frac{1}{2}(I+A) \Rightarrow P^2 = \frac{1}{4}(I+2A+A^2) = \frac{1}{2}(I+A) = P$$

use $A^2=I$; same for $Q=\frac{1}{2}(I-A)$

Memory hook Idempotent = “নিজেকে square করেও unchanged”; Involutory = “square করলে identity”; Nilpotent = “power বাড়াতে বাড়াতে zero”.

5. Hermitian and skew-Hermitian

$$A^\dagger = (\text{conjugate of } A)^T \quad | \quad \text{Hermitian: } A^\dagger=A \quad | \quad \text{Skew-Hermitian: } A^\dagger=-A$$

Real entries হলে conjugate বদলায় না, তাই real matrix-এর ক্ষেত্রে Hermitian becomes symmetric, এবং skew-Hermitian becomes skew-symmetric. Hermitian matrix-এর diagonal entries real; skew-Hermitian-এর diagonal entries purely imaginary or zero.

$$\text{Complex decomposition: } A = \frac{1}{2}(A+A^\dagger) + \frac{1}{2}(A-A^\dagger)$$

first part Hermitian, second part skew-Hermitian

6. Matrix arithmetic is not ordinary real-number arithmetic

Property that can fail	Counterexample / reason
$AB=BA$	Generally false; matrix multiplication is non-commutative.
$AB=0 \Rightarrow A=0$ or $B=0$	False: nonzero matrices can be zero divisors.
$AB=AC \Rightarrow B=C$	False if A is singular; cancellation needs suitable invertibility.

Zero-divisor example: $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ are both nonzero but $AB=0$. Cancellation failure: with $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $B=I$ and $C = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$, $AB=AC$ although $B \neq C$.

7. Proof mini-bank for A1

Proof / derivation	One-line route
Diagonal entries of skew-symmetric are zero	$a_{ii} = -a_{ii} \Rightarrow 2a_{ii}=0$
If A symmetric & nonsingular, A^{-1} symmetric	$(A^{-1})^T = (A^T)^{-1} = A^{-1}$

Proof / derivation	One-line route
$A=P+Q$ symmetric + skew, uniquely	$P=\frac{1}{2}(A+A^T)$, $Q=\frac{1}{2}(A-A^T)$
Involutory $\Rightarrow \frac{1}{2}(I \pm A)$ idempotent	expand square and use $A^2=I$
Direction-cosine matrix is orthogonal	rows/columns are orthonormal $\Rightarrow AA^T=I$

Matching questions from the question bank — A1

Year / Q	Matching question family	Priority
2015 Q1	Equivalent, singular, trace, symmetric; matrix arithmetic exceptions	★★★
2016 Q1 / Q4	Matrix/scalar/Hermitian/trace; Hermitian decomposition; symmetric inverse	★★★
2017 Q5	Skew-symmetric, principal submatrix, elementary, diagonal; decomposition	★★★
2018 Q1-Q4	Orthogonal/Hermitian/idempotent/lower triangular; decomposition; skew diagonal zero; involutory proof	★★★
2019 Q1/Q3	Orthogonal test; symmetric parameter problem	★★
2021 Q4	Skew-symmetric, singular, diagonal definitions	★★
2022 Q7	Direction-cosine matrix $\rightarrow$ orthogonal proof	★★
2023 Q5	Skew-symmetric/diagonal/orthogonal + decomposition	★★★
2024 Q5	Orthogonal/elementary/non-singular definitions	★★★
2025 Q1	Lower triangular/symmetric/nilpotent/idempotent; nilpotent index; decomposition	★★★

A1 rapid recall - 5 minute self-test

Without looking (1) Write conditions for symmetric, skew-symmetric, orthogonal, idempotent, nilpotent. (2) Prove skew-symmetric diagonal = 0. (3) Write $A=P+Q$ decomposition. (4) State three ways matrices differ from real-number arithmetic.

VERY HIGH PRIORITY

SET A2 — Inverse, Rank, Consistency & Linear Dependence

এই set-এর সবচেয়ে বড় advantage: নিয়ম ঠিক থাকলে answer প্রায় mechanical. ভুল কমানোর মূল কৌশল হলো row operation পরিষ্কার লিখে প্রতিটি step audit করা।

1. Elementary transformations (ERO/ECO)

Operation	Row form	What it does
Interchange	$R_i \leftrightarrow R_j$	row swap
Scaling	$R_i \rightarrow kR_i, k \neq 0$	pivot normalize
Replacement	$R_i \rightarrow R_i + kR_j$	eliminate an entry

Exam discipline যে operation করবেন, left margin-এ লিখুন। এক line-এ অনেক operation করলে sign error ধরা কঠিন হয়।

2. Inverse by Gauss-Jordan elementary row transformation

$$[A \mid I] \xrightarrow{\text{same row operations}} [I \mid A^{-1}]$$

possible only when A is nonsingular / full rank

Invertibility quick test for $n \times n$ square A: $\det(A) \neq 0 \Rightarrow \text{rank}(A) = n \Rightarrow A^{-1}$ exists. এই তিনটি condition-কে D-R-I হিসেবে মনে রাখুন: Determinant, Rank, Inverse.

Shortcut — D-R-I $\det \neq 0 \rightarrow$ full rank $\rightarrow$ inverse exists. $\det = 0 \rightarrow \text{rank} < n \rightarrow$ inverse does not exist.

Worked class-note example

$$A = \begin{bmatrix} -2 & 1 & 3 \\ 0 & -1 & 1 \\ 1 & 2 & 0 \end{bmatrix}, \quad \det(A) = 8 \neq 0$$

Augment with identity and row-reduce. The class-note Gauss-Jordan sequence yields:

$$A^{-1} = \begin{bmatrix} -1/4 & 3/4 & 1/2 \\ 1/8 & -3/8 & 1/4 \\ 1/8 & 5/8 & 1/4 \end{bmatrix}$$

Final check $A \cdot A^{-1} = I$ (or $A^{-1} \cdot A = I$) লিখে verify করুন। বিশেষ করে fraction-heavy inverse-এ এই check marks বাঁচায়।

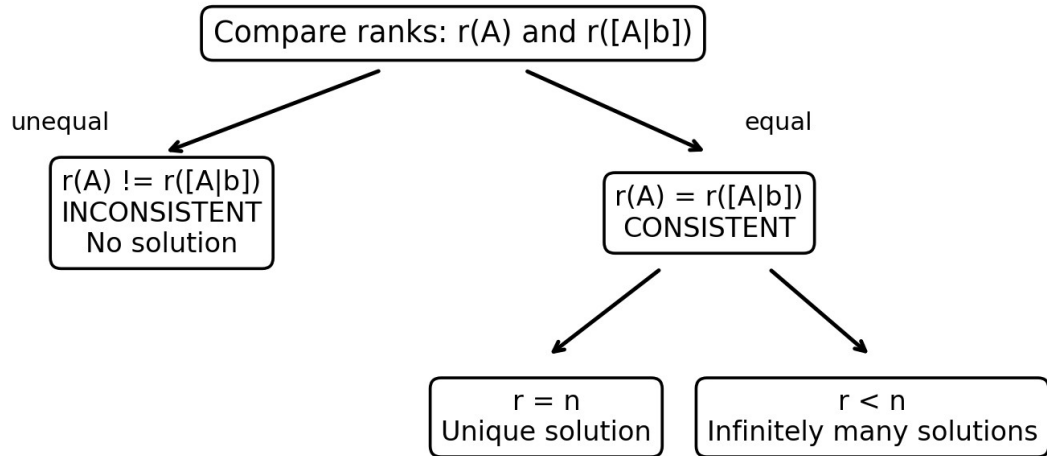
3. Rank — “কতগুলো independent information আছে?”

Rank is the maximum number of linearly independent rows/columns. Row-reduction-এ rank = echelon form-এর nonzero rows/pivots-এর সংখ্যা।

$$\text{rank}(A) = \text{number of pivots} = \text{number of nonzero rows in REF}$$

Important: elementary row/column operations rank পরিবর্তন করে না। Normal form $[I_r \ 0; \ 0 \ 0]$ -এ identity block-এর order r -ই rank.

4. Consistent / inconsistent system — Rouché-Capelli rank rule



Rank rule: first compare two ranks, then compare the common rank with number of unknowns n .

Decision tree for $Ax=b$ using $\text{rank}(A)$, $\text{rank}([A|b])$ and number of unknowns n .

Shortcut — R-R-n Step 1: $\text{rank}(A)$ বনাম $\text{rank}([A|b])$ compare. Step 2: equal হলে common rank r বনাম unknown count n compare.

Condition	Conclusion	বাংলা meaning
$r(A) \neq r([A b])$	Inconsistent	সমাধান নেই
$r(A)=r([A b])=n$	Unique solution	একটি সমাধান
$r(A)=r([A b])<n$	Infinitely many solutions	অসীম সমাধান / free variable আছে

Parameter technique example

System: $x+y+z=1$; $2x+3y+4z=2$; $x+2y+3z=k$. Coefficient-wise, row 2 = row 1 + row 3. Therefore consistency demands $\text{RHS } 2 = 1+k$, so $k=1$.

If $k=1$, $\text{rank}(A)=\text{rank}([A|b])=2<3 \rightarrow$ infinitely many solutions. If $k \neq 1$, augmented rank becomes larger $\rightarrow$ inconsistent.

Fast insight Parameter problem-এ আগে coefficient rows-এর relation খুঁজুন। RHS একই relation মানে কিনা দেখলে অনেক সময় full elimination ছাড়াই condition পাওয়া যায়।

5. Homogeneous system $Ax=0$: trivial vs non-trivial

$Ax = 0$ always has $x=0$ (trivial solution).

Case	Square A	General $m \times n$ viewpoint
Only trivial	$\det(A) \neq 0$	$\text{rank}(A) = n$ (no free variable)
Non-trivial exists	$\det(A) = 0$	$\text{rank}(A) < n$ (at least one free variable)

বাংলা cue Trivial = সব variable zero. Non-trivial = অন্তত একটি variable nonzero.

6. Linear dependence / independence

Vectors $v_1, \dots, v_n$ are linearly independent if $c_1 v_1 + c_2 v_2 + \dots + c_n v_n = 0$ only has $c_1 = c_2 = \dots = c_n = 0$. Otherwise they are dependent.

Put vectors as columns of M. If $\text{rank}(M) = \text{number of vectors} \rightarrow$ independent; otherwise dependent.

Example: $v_1 = (1, 2, 1)$, $v_2 = (2, 1, 3)$, $v_3 = (3, 3, 4)$. Since $v_3 = v_1 + v_2$, the set is dependent immediately — no determinant needed.

Shortcut প্রথমে obvious relation খুঁজুন (sum, multiple, zero vector). না পেলো rank/determinant method নিন।

7. Inverse existence checklist

Equivalent statement for square A	Exam use
$\det(A) \neq 0$	fastest numerical test
$\text{rank}(A) = n$	row-reduction context
$Ax = 0$ has only trivial solution	systems context
columns/rows are linearly independent	vector context
A row-reduces to I	Gauss-Jordan context

Matching questions from the question bank — A2

Year / Q	Matching question family	Priority
2015 Q2-Q4	Inverse by ERO; parameter consistency; homogeneous non-trivial; vector dependence	★★★
2016 Q2-Q3	Inverse/existence; systems; trivial vs non-trivial	★★★
2017 Q6	Inverse by ERO + consistency/system classification	★★★
2018 Q1/Q3	Inverse by ERO + consistency / many solutions	★★★
2019 Q1-Q2	Parameter solution; parameter-dependent rank; inverse work	★★★
2020 Q3/Q6	Augmented system, consistency, rank/adjoint	★★
2021 Q4 / Sec-B system	Inverse by ERO + linear systems	★★
2022 Q4	Parameter-dependent solution type	★★★
2023 Q5-Q6	Inverse/product of elementary matrices; dependence; homogeneous/system parameters	★★★
2024 Q5-Q6	Dependence; invertibility/inverse; parameter system	★★★
2025 Q2	Inverse existence + ERO; parameter system; dependence	★★★

VERY HIGH PRIORITY

SET A3 — Echelon / Canonical / Normal Form + Eigen Block

এই set-এ দুইটি recurring engine আছে: (i) row/column reduction ও rank, (ii) characteristic equation থেকে eigenvalue/eigenvector. Step-order ঠিক রাখাই মূল।

1. REF, RREF (row canonical) and normal form

Form	Core condition	Use
REF / Echelon	nonzero rows above zero rows; pivots staircase to the right; entries below pivots zero	rank + back substitution
RREF / Row canonical	REF + each pivot is 1 and the only nonzero entry in its column	canonical row solution form
Normal form (equivalence)	using row + column ops: $[I_r \ 0; \ 0 \ 0]$	rank + equivalence

Accuracy note Some class slides normalize REF pivots to 1. Standard REF only requires a nonzero leading entry; normalizing it to 1 is allowed and convenient. RREF, however, definitely requires pivot 1 and zeros above/below each pivot.

2. Reduction workflow

1. Create first pivot; clear entries below it.
2. Move to next row/column; repeat staircase elimination $\rightarrow$ REF.
3. Normalize pivots to 1 and clear above pivots $\rightarrow$ RREF / row canonical form.
4. If normal form is required, use column operations too to convert RREF to $[I_r \ 0; \ 0 \ 0]$.
5. Rank = number of pivots = r .

Small worked example

$$E = [[1, 2, 1], [2, 4, 2], [1, 1, 0]]$$

$$\text{RREF}(E) = [[1, 0, -1], [0, 1, 1], [0, 0, 0]], \quad \text{rank}(E) = 2$$

Then column operation $C_3 \rightarrow C_3 + C_1 - C_2$ gives normal form $\text{diag}(1, 1, 0)$.

Normal-form cue Normal form দেখলেই block identity চিনুন: identity block-এর size = rank.

3. Tracking P and Q in PAQ = normal form

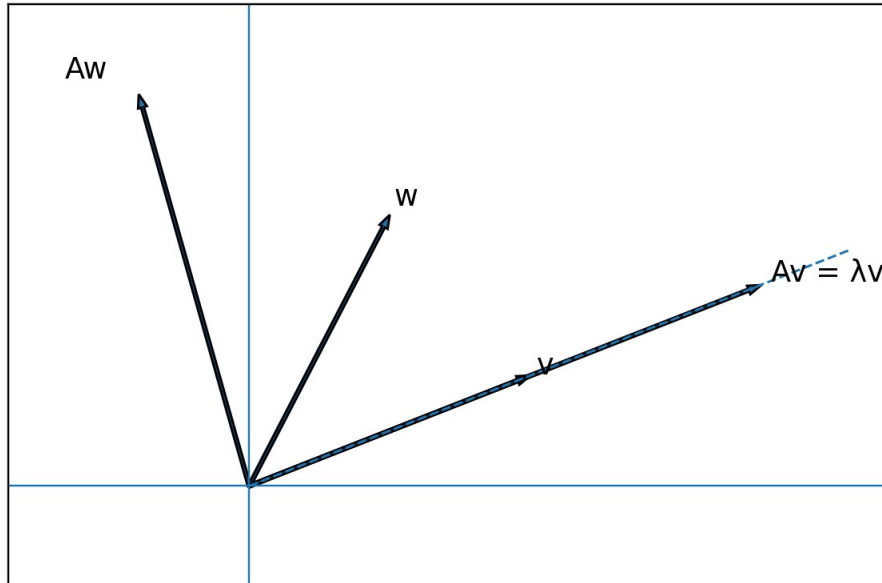
If a problem asks for non-singular P, Q such that $PAQ = N$, keep a “shadow identity” beside A: every row operation on A is applied to the left identity to build P; every column operation is applied to the right identity to build Q.

$$P A Q = N = [[I_r, 0], [0, 0]]$$

Common mistake Row operations generate left multiplication ($P \cdot A$). Column operations generate right multiplication ($A \cdot Q$). এই দিক উল্টে ফেলবেন না।

4. Eigenvalue and eigenvector — intuition

Eigenvector keeps its line/direction; a general vector usually changes direction



Eigenvector একটি বিশেষ direction: A প্রয়োগ করলে vector সেই একই line-এ থাকে; শুধু scale (এবং $\lambda < 0$ হলে direction sense) বদলায়।

$$A x = \lambda x, \quad x \neq 0$$

বাংলা **cue** Eigenvector = transformation-এর পর "দিকের line একই"; eigenvalue λ = কত গুণ stretch/shrink (বা sign flip) হলো।

5. D-S-V algorithm for eigen problems

1. D — Determinant: form $\det(A - \lambda I) = 0$ (or monic $\det(\lambda I - A) = 0$).
2. S — Solve the characteristic equation for λ .
3. V — Vector: for each λ solve $(A - \lambda I)x = 0$ and take any nonzero solution.

Sign convention $\det(A - \lambda I) = 0$ এবং $\det(\lambda I - A) = 0$ roots একই দেয় (global sign may differ for odd order). একটি convention ধরে শেষ পর্যন্ত consistent থাকুন।

Worked class-note example

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$$

Characteristic equation gives eigenvalues $\lambda = -2, 3, 6$. Corresponding convenient eigenvectors are:

λ	One eigenvector x	Quick check
-2	$(-1, 0, 1)^T$	$Ax = -2x$

λ	One eigenvector x	Quick check
3	$(1, -1, 1)^T$	$Ax=3x$
6	$(1, 2, 1)^T$	$Ax=6x$

Eigenvector freedom If x is an eigenvector, any nonzero scalar multiple cx is also an eigenvector for the same λ . তাই answer অন্য multiple হলেও correct.

6. High-value eigen properties

Property	Formula / statement
Trace	Σ eigenvalues = $\text{tr}(A)$, counting algebraic multiplicity
Determinant	Π eigenvalues = $\det(A)$
Triangular matrix	eigenvalues are diagonal entries
Transpose	A and A^T have the same characteristic polynomial/eigenvalues
Inverse	if A invertible: eigenvalues of A^{-1} are $1/\lambda$
Power	if $Ax=\lambda x$, then $A^m x = \lambda^m x$
Distinct λ	eigenvectors from distinct eigenvalues are linearly independent
Real symmetric A	eigenvalues real; eigenvectors for distinct λ are orthogonal

7. Diagonalization

$$A = P D P^{-1} \Leftrightarrow P^{-1} A P = D$$

columns of P are independent eigenvectors; D carries matching eigenvalues

An $n \times n$ matrix is diagonalizable if it has n linearly independent eigenvectors.

Worked 2x2 example

$$H = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}; \quad \lambda_1=5 \text{ with } v_1=(1,1)^T; \quad \lambda_2=2 \text{ with } v_2=(-1,2)^T$$

$$P = \begin{bmatrix} 1 & -1 \\ 1 & 2 \end{bmatrix}, \quad D = \text{diag}(5, 2), \quad P^{-1} H P = D$$

Ordering rule P -এর column order এবং D -এর diagonal eigenvalue order অবশ্যই match করবে।

Matching questions from the question bank — A3

Year / Q	Matching question family	Priority
2015 Q2-Q4	Echelon $\rightarrow$ canonical $\rightarrow$ normal + rank; eigen/large eigen; eigen properties	★★★
2016 Q2/Q4	Row-reduced/normal/rank; eigenvalues/eigenvectors	★★★
2017 Q5/Q7-Q8	Forms/rank; eigen; find P, Q for normal form	★★★
2018 Q2/Q4	Forms, P, Q ; eigen block	★★★
2019 Q2-Q4	Eigen / diagonalization / characteristic equation variants	★★★

Year / Q	Matching question family	Priority
2020 Q3/Q6	Eigen + rank	★★
2021 Q7	Eigenvalues/eigenvectors	★★
2022 Q4/Q7	Normal/rank + largest eigen	★★★
2023 Q6-Q7	Normal/PQ + inverse-eigenvalue properties	★★★
2024 Q6-Q7	Normal/PQ + eigen block	★★★
2025 Q3	Echelon/canonical/normal + P,Q + eigenvalues/eigenvectors	★★★

A3 rapid self-test

Task	You should be able to do it in
Convert a 3x3 matrix to REF/RREF and state rank	4-6 min
Recognize / produce normal form $[I_r \ 0; 0 \ 0]$	2 min
Find eigenvalues then one eigenvector for each λ	6-8 min
Build P,D for a diagonalizable 2x2 matrix	4 min

D-S-V check Determinant $\rightarrow$ Solve $\lambda \rightarrow$ Vector. If this sequence is automatic, most eigen questions become routine.

MEDIUM-HIGH PRIORITY / STRONG BACKUP

SET A4 — Vector Kinematics, Work Done & Cayley-Hamilton

Set A4 formula-driven. Vector motion-এর diagram/geometry বুঝে নিলে memorization অনেক কমে যায়। Work done-এ parameterization এবং Cayley-Hamilton-এ characteristic polynomial-এর discipline সবচেয়ে গুরুত্বপূর্ণ।

1. Position, velocity, speed and acceleration

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k} \quad | \quad \mathbf{v} = \mathbf{r}'(t) \quad | \quad \text{speed} = |\mathbf{v}| \quad | \quad \mathbf{a} = \mathbf{r}''(t)$$

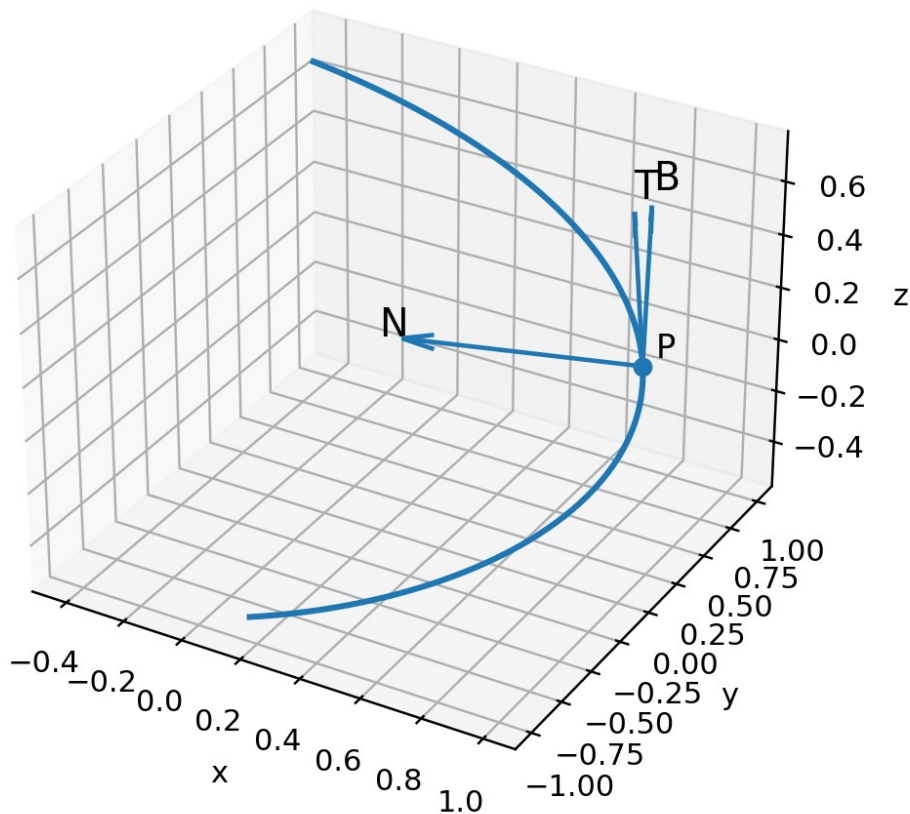
বাংলা cue Position $\mathbf{r}$ = “কোথায় আছি”; velocity $\mathbf{v}$ = “কোন দিকে কত দ্রুত যাচ্ছি”; acceleration $\mathbf{a}$ = “velocity কত দ্রুত বদলাচ্ছে”.

Velocity vector tangent to the path. Acceleration need not be tangent; it can have tangential and normal components.

$$\mathbf{a} = a_T \mathbf{T} + a_N \mathbf{N}, \quad a_T = d|\mathbf{v}|/dt, \quad a_N = \kappa |\mathbf{v}|^2$$

2. Frenet frame: T, N, B

Frenet trihedron: tangent T, principal normal N, binormal B



At a regular point P , T follows the curve, N points toward bending, $B = T \times N$ completes the perpendicular frame.

$$\mathbf{T} = \mathbf{r}'/|\mathbf{r}'| \quad | \quad \mathbf{B} = (\mathbf{r}' \times \mathbf{r}'')/|\mathbf{r}' \times \mathbf{r}''| \quad | \quad \mathbf{N} = \mathbf{B} \times \mathbf{T}$$

Memory hook $\mathbf{v} \rightarrow \mathbf{T}$, then bending gives $\mathbf{N}$, and $\mathbf{T} \times \mathbf{N} \rightarrow \mathbf{B}$. তিনটি unit vector পরস্পর perpendicular.

3. Curvature, torsion and the three planes

$$\kappa = |\mathbf{r}' \times \mathbf{r}''| / |\mathbf{r}'|^3 \quad | \quad \tau = ((\mathbf{r}' \times \mathbf{r}'') \cdot \mathbf{r}''') / |\mathbf{r}' \times \mathbf{r}''|^2$$

Object	Normal vector	Equation at point $\mathbf{r}_0$
Normal plane	$\mathbf{T}$	$\mathbf{T} \cdot (\mathbf{r} - \mathbf{r}_0) = 0$
Osculating plane	$\mathbf{B}$	$\mathbf{B} \cdot (\mathbf{r} - \mathbf{r}_0) = 0$
Rectifying plane	$\mathbf{N}$	$\mathbf{N} \cdot (\mathbf{r} - \mathbf{r}_0) = 0$

বাংলা **meaning** Curvature κ = curve কত তীব্র বাঁকছে। Torsion τ = curve space-এ osculating plane থেকে কত দ্রুত মোচড়াচ্ছে।

Worked reference-note example at $t=2$

Curve: $\mathbf{r}(t) = (t^2-1)\mathbf{i} + (4t-3)\mathbf{j} + (2t^2-6t)\mathbf{k}$. At $t=2$: $\mathbf{r}_0 = (3, 5, -4)$, $\mathbf{v} = (4, 4, 2)$, $|\mathbf{v}| = 6$.

$$\mathbf{T} = (2, 2, 1)/3 \quad | \quad \mathbf{N} = (1, -8, 14)/(3\sqrt{29}) \quad | \quad \mathbf{B} = (4, -3, -2)/\sqrt{29}$$

$$\kappa = \sqrt{29}/54, \quad \tau = 0, \quad \text{radius of curvature } \rho = 54/\sqrt{29}$$

$$\text{Normal plane: } 2x + 2y + z - 12 = 0$$

$$\text{Osculating plane: } 4x - 3y - 2z - 5 = 0$$

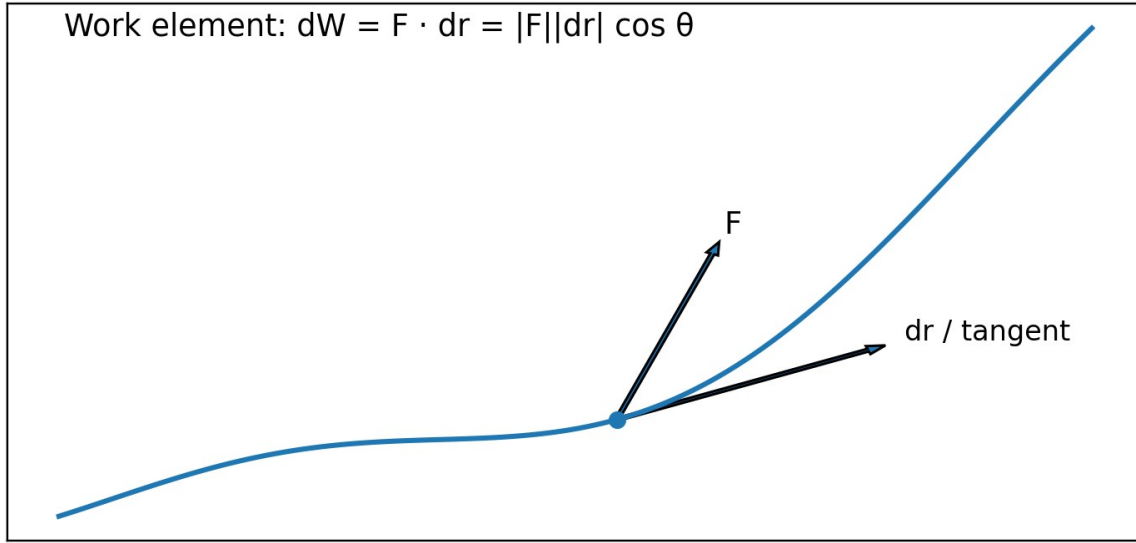
$$\text{Rectifying plane: } x - 8y + 14z + 93 = 0$$

Interpretation For this quadratic curve, $d^3\mathbf{r}/dt^3 = 0$; therefore $\tau=0$ throughout the regular part. Hence the curve is planar and its osculating plane is fixed.

4. Work done by a force field

Work along a curve: only the tangential component of force contributes

Work element: $dW = \mathbf{F} \cdot d\mathbf{r} = |\mathbf{F}||d\mathbf{r}| \cos \theta$



F-এর tangent-direction component-টাই কাজ করে; তাই dot product naturally appears.

$$W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int [P dx + Q dy + R dz]$$

1. Choose one parameter (t or x) for the entire path.
2. Write x,y,z and dx,dy,dz in that parameter.
3. Substitute into $\mathbf{F} \cdot d\mathbf{r}$.
4. Use correct parameter limits and integrate.
5. For a piecewise path, calculate each segment separately and add.

Repeated KUET question pattern — 2016 & 2017

$\mathbf{A} = (2y+3)\mathbf{i} + xz\mathbf{j} + (yz-x)\mathbf{k}$. Evaluate $\int \mathbf{A} \cdot d\mathbf{r}$ (i) along $O \rightarrow (0,0,1) \rightarrow (0,1,1) \rightarrow (2,1,1)$, and (ii) along the straight line $O \rightarrow (2,1,1)$.

Path	Parameter idea	Answer
Piecewise	three segments; two give 0, last gives $\int_0^2 5dx$	10
Straight	$x=2t, y=t, z=t, 0 \leq t \leq 1$	8

What this teaches Same endpoints হলেও field conservative না হলে path বদলালে work বদলাতে পারে. এখানে $10 \neq 8$, so path dependence is visible.

Another repeated family — 2020 & 2024

$\mathbf{F} = 3x^2\mathbf{i} + (2xz-y)\mathbf{j} + z\mathbf{k}$, along $x^2=4y$ and $3x^3=8z$ from $x=0$ to 2 . Parameterize by x ; the compiled question solution gives $W=16$.

5. Force $\rightarrow$ motion

$$\mathbf{F} = m \mathbf{a} = m \mathbf{r}''(t)$$

If force $F(t)$ is known, integrate acceleration to get velocity, then integrate velocity to get position. Each integration introduces a constant fixed by initial conditions $r(t_0)$, $v(t_0)$.

Shortcut $F \rightarrow a \rightarrow v \rightarrow r$. Arrow direction is "integrate twice"; going backward $r \rightarrow v \rightarrow a$ means differentiate.

6. Cayley-Hamilton theorem



Shortcut: characteristic polynomial $\rightarrow$ matrix equation $\rightarrow$ lower powers / inverse

Cayley-Hamilton workflow: turn the scalar characteristic equation into a matrix identity.

If $p(\lambda) = \det(\lambda I - A)$, then every square matrix A satisfies $p(A) = 0$.

Main uses in this course: (i) verify that a matrix satisfies its characteristic equation, (ii) reduce high powers A^n , (iii) find A^{-1} when $\det(A) \neq 0$.

Worked inverse example

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad p(\lambda) = \lambda^2 - 5\lambda - 2$$

$$\text{By C-H: } A^2 - 5A - 2I = 0. \text{ Multiply by } A^{-1}: A - 5I - 2A^{-1} = 0.$$

$$A^{-1} = \frac{1}{2}(A - 5I) = \begin{bmatrix} -2 & 1 \\ 3/2 & -1/2 \end{bmatrix}$$

CH shortcut Inverse বের করতে characteristic equation-এ constant term $c_0 \neq 0$ হলে terms rearrange করে A^{-1} isolate করুন. High power হলে equation repeatedly use করে power কমান।

Matching questions from the question bank — A4

Year / Q	Matching question family	Priority
2015 Q5	Particle velocity/acceleration + vector motion concepts	★★
2016 Q7 / 2017 Q4	Repeated line/work integral along piecewise vs direct path	★★★
2018 vector set	Tangent/force/work and scalar triple product families	★★
2020 Q1-Q2	Vector derivative/work; work field later repeats in 2024	★★★
2021 Q1-Q2	Moving particle, T/N/B/curvature-type geometry + work	★★★
2022 vector section	Directional/vector geometry; work/integration families	★★
2023 Q8 / vector block	Particle velocity/acceleration and vector curve questions	★★
2024 Q2 + CH block	Work question repeats 2020; Cayley-Hamilton verify/use	★★★
2025 Q4	Particle motion, work, Cayley-Hamilton theorem/inverse family	★★★

A4 rapid self-test

Formula sprint From memory write: $\mathbf{v}=\mathbf{r}'(t)$, $\mathbf{a}=\mathbf{r}''(t)$, T , B , N , curvature κ , torsion τ , work $W=\int \mathbf{F} \cdot d\mathbf{r}$, and the Cayley-Hamilton statement. Then solve one parameterized work integral without looking at the note.

FINAL REVISION

SECTION A — One-page Formula & Decision Sheet

Block	Must-remember formulas / rules
Matrix types	$A^T=A$ symmetric; $A^T=-A$ skew; $A^T A=I$ orthogonal; $A^2=A$ idempotent; $A^2=I$ involutory; $A^k=0$ nilpotent
Decomposition	$A=\frac{1}{2}(A+A^T)+\frac{1}{2}(A-A^T)$; complex version uses $A^\dagger$
Invertible	$\det A \neq 0 \Rightarrow \text{rank } A=n \Rightarrow A^{-1}$ exists $\Rightarrow Ax=0$ only trivial
Inverse ERO	$[A I] \rightarrow [I A^{-1}]$
Consistency	$r(A) \neq r([A b])$ no solution; equal rank= n unique; equal rank $<n$ infinite
Forms	REF staircase; RREF pivot 1 & pivot column cleared; normal= $[I_r \ 0 \ 0 \ 0]$
Eigen	$\det(A-\lambda I)=0$; then $(A-\lambda I)x=0$; diagonalize $A=PDP^{-1}$
Vector motion	$v=r'$, $a=r''$, $T=v/ v $, $B=(r' \times r'')/ r' \times r'' $, $N=B \times T$
Curve geometry	$\kappa= r' \times r'' / r' ^3$; $\tau=((r' \times r'') \cdot r''')/ r' \times r'' ^2$
Work	$W=\int F \cdot dr$
Cayley-Hamilton	$p(\lambda)=\det(\lambda I-A) \Rightarrow p(A)=0$

Proof / derivation priority list

Priority	Proof / derivation	Why
1	$A = \text{symmetric} + \text{skew-symmetric}$, uniqueness	Repeated and easy marks
2	skew-symmetric diagonal entries are zero	Very short proof
3	symmetric nonsingular $\Rightarrow$ inverse symmetric	One identity proof
4	involutory $\Rightarrow \frac{1}{2}(I \pm A)$ idempotent	Repeated expansion pattern
5	direction-cosine matrix orthogonal	Orthogonality proof
6	Cayley-Hamilton verify + use for inverse	Longer, high-mark backup

Common mistakes — last-minute warning

- Equivalent vs row-equivalent: full equivalence may use row + column operations ($B=PAQ$); row-equivalence uses row operations only ($B=PA$).
- RREF/canonical and normal form are not the same object. Normal form under equivalence is $[I_r \ 0 \ 0 \ 0]$.
- In eigen problems, $x=0$ is never an eigenvector.
- For diagonalization, keep eigenvector column order in P aligned with eigenvalue order in D .
- In consistency problems, never decide unique/infinite before first checking $r(A)=r([A|b])$.
- In inverse problems, row operations must be applied to the whole augmented matrix $[A|I]$.
- In line integral, change dx, dy, dz together with x, y, z after parameterization.
- In TNB problems, normalize carefully; B is based on $r' \times r''$ and $N=B \times T$ for the convention used here.
- Cayley-Hamilton: use one characteristic-polynomial sign convention consistently.

3-out-of-4 practical decision plan

If exam paper looks like...	Recommended 3 sets
Standard / balanced	A1 + A2 + A3
A3 eigen algebra is unusually ugly	A1 + A2 + A4
A1 theory contains unfamiliar niche definitions	A2 + A3 + A4
A2 parameter system is long but A4 is familiar	A1 + A3 + A4

Best default Prepare A1-A3 to full confidence and A4 to at least 70%. That gives you an escape route without weakening the three most repeated blocks.

Source basis used for this class note

- Matrix Lec 01: types of matrices, transpose, elementary/equivalent matrices, trace and related examples.
- Matrix Lec 02: elementary row/column operations, echelon, row-canonical/RREF and normal form.
- Matrix Lec 03-04: consistent/inconsistent systems, rank criteria and dependence/independence.
- Matrix Lec 05: inverse by elementary transformation / Gauss-Jordan.
- Matrix Lec 07: eigenvalue/eigenvector definition and worked problems.
- Uploaded KUET Math 2123 term-question collection and 2015-2024 topic compilations for matching-question tags.
- Uploaded H.K. Dass / matrix reference books for formula checks and supporting derivations.

Made by 2223033